

# Fiona Victoria Stanley Jothiraj

Beaverton, OR

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## EDUCATION

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### Oregon State University, Corvallis

2023 — 2027

*Doctor of Philosophy (PhD) in Artificial Intelligence*

GPA: 3.79/4.00

Area of Research: AI for Social Good, Applied AI/ML in Ecology

Honors: Outstanding Scholars Program

### University of Washington Bothell

2021 — 2023

*Master of Science in Computer Science and Software Engineering*

GPA: 3.90/4.00

Thesis: Phoenix - A Federated Generative Diffusion Model

### PSG College of Technology, India

2015 — 2019

*Bachelor of Engineering in Robotics and Automation Engineering*

GPA: 9.58/10.00

## RESEARCH EXPERIENCE

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### Graduate Researcher

Oct 2023 - present

Oregon State University

*Advisor: Dr. Rebecca Hutchinson*

- Modeled plant-pollinator interactions across a decade of environmental change using Bayesian hierarchical dynamic occupancy models and probabilistic graphical models in R
- Investigated imperfect detection effects in species distribution models by simulating detection noise across linear and hierarchical statistical frameworks
- Conducted interdisciplinary research on species distribution modeling, demonstrating a 23% performance gain with boosted regression trees (BRTs) when habitat sampling is centered on bird locations rather than surveyor locations

### AI Research Intern

June 2024/25 - Sep 2024/25

Micron Technology

*Advisor: Seth Eichmeyer, Director of Product Yield Analytics & Analysis*

- Awarded the **Micron Innovation Award**, marking the first step toward a patent filing
- Designed and deployed ML models to optimize semiconductor product yield through wafer defect analysis
- Built a non-parametric clustering system with rotation- and flip-invariant features, reducing manual analysis time by 90% (*AAAI'26 Workshop & Bridge Program*)
- Developed a novel anomaly scoring metric using conditional VAEs, diffusion models, and ViTs to detect multi-defect patterns in semiconductor manufacturing
- Presented project outcomes and business impact to the VP of Engineering

### Graduate Researcher

June 2022 - Jan 2023

UW Bothell: Computation Behavioral Modeling (CBM) Research Lab

*Advisor: Dr. Afra Mashhadi*

- Conducted research on nostalgic behavior on social media using NLP and LLMs
- Built transformer models, including RoBERTa, DistilBERT, and an ensemble-based model for nostalgic tweet detection, achieving 0.96 Micro F1-Score
- Leveraged LLaMA2-70B to curate a large-scale dataset of  $\approx 250K$  nostalgic English tweets

## PUBLICATIONS

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- [1] **Stanley Jothiraj, Fiona Victoria**, Hutchinson Rebecca, Lauren Ponisio, Julia Jones, Nicole Martinez (2026). Plant-Pollinator Interactions over more than a Decade of Environmental Change. *under review*
- [2] **Fiona Victoria Stanley Jothiraj**, Arunagiri Pandian, Seth A. Eichmeyer. (2026) DECOR: Deep Embedding Clustering with Orientation Robustness. *AAAI 2026 Bridge Program on Knowledge-guided Machine Learning*. Also selected at the *AAAI 2026 Workshop on AI to Accelerate Science and Engineering (AI2ASE)*
- [3] Shen Fang-Yu, **Stanley Jothiraj, Fiona Victoria**, Hutchinson Rebecca A, Hallman Tyler, Curtis Jenna R, and Robinson, W. Douglas. (2025). Species Distribution Model Performance Improves When Habitat

Characterizations are Centered on Detected Individuals Instead of Observers. *Ecological Indicators*

- [4] **Stanley Jothiraj, F. V.**, & Mashhadi, A. (2024). Phoenix: A Federated Generative Diffusion Model. *Companion Proceedings of the ACM on Web Conference 2024 (WWW 2024)*
- [5] **Stanley Jothiraj, F. V.**, Hong L., & Mashhadi, A. (2024) Nostalgia on Twitter: Detection and Analysis of a Large-Scale Dataset. *Proceedings of the Association for Information Science and Technology (ASIS&T)*
- [6] Fang-Yu Shen, **Fiona Victoria Stanley Jothiraj**, Tyler A. Hallman, Rebecca A. Hutchinson, W. Douglas Robinson. (2024). Does species distribution model performance improve when habitat sampling is centered on bird locations instead of surveyor locations? *American Ornithological Society Annual Meeting*
- [7] **Stanley Jothiraj, F. V.** (2022). Time Series Prediction for Food Sustainability. *arXiv preprint*
- [8] Shamsaddini, Vahid, **Stanley Jothiraj, Fiona Victoria**, Chen, Mandy, & Mashhadi, Afra. (2022). Empirical Dynamic Modelling of the Multi-Source Park Visitation Data. *Data for Policy Conference*

## INVITED TALKS

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|                                         |                                                         |      |
|-----------------------------------------|---------------------------------------------------------|------|
| • Ai4 Conference                        | Solo Speaker @ Edge AI & Tiny ML Technical Track        | 2026 |
| • ICCV CVAM Workshop                    | Creativity and Privacy in Advertising & Marketing       | 2025 |
| • Decentralized AI Hackathon @ Stanford | Federated Learning for Generative Models                | 2025 |
| • University of Washington              | Federated Learning for Generative Diffusion Models      | 2025 |
| • Flower Labs                           | Phoenix: The First Federated Generative Diffusion Model | 2023 |

## EXPERIENCE

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### Graduate Teaching Assistant

Jan 2026 - June 2026

Oregon State University

- Courses: CS 464 Open Source Software & CS 493 Cloud Application Development
- Supported over 300 students in mastering open source workflows and cloud-native development practices
- Delivered feedback on assignments and led office hours to reinforce hands-on technical concepts

### Data Scientist

June 2023 - Oct 2023

Harvard in Tech

- Led NLP research as a thought leader in the Call for Action (CFA) team to detect unreliable news articles
- Influenced executive leadership to shape the roadmap of the CFA Data Science team
- Built transformer-based models to classify news headlines using open-source and in-house curated data
- Presented findings on fake news detection, feature engineering, and state-of-the-art NLP approaches

### Machine Learning Engineer

June 2019 - Mar 2020

Multicoreware Inc, India

- Designed and deployed EV Quantization logic in TensorFlow GPU for quantization-aware training and Tensorflow Lite inference of Deep Neural Networks
- Shipped open-source product to production powering Synopsys DesignWare EV Processors
- Mentored peers on quantization concepts and EV TensorFlow workflows using C++, Python, and Intel Intrinsics

### Machine Learning Intern

Dec 2018 - May 2019

Multicoreware Inc, India

- Developed a custom CNN model for skin cancer detection by training on gigabytes of clinical image data
- Implemented the LipSync™ API in Intel's OpenVINO via a high-level C++ inference engine for 5x speedup
- Deployed the LipSync quality control tool for OTT streaming providers using deep learning
- Demonstrated LipSync™ technology at the National Association of Broadcasters Show (NAB 2019)

## AWARDS

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|-----------------------------------------------------------------------------------------------------|-----------|
| • <b>Professional Development Award</b> - Oregon State University                                   | 2024/2025 |
| • <b>Micron Innovation Award</b> - Micron Technology                                                | 2024      |
| • <b>Outstanding Scholars Program</b> - Oregon State University                                     | 2023      |
| • <b>Virtual Scholarship</b> - Grace Hopper Women in Computing Celebration                          | 2022      |
| • <b>Academic Excellence</b> (Ranked 1/80) - Robotics and Automation Engineering Association        | 2019      |
| • <b>Monarch of the Month</b> - Individual contribution to TensorFlow quantization at Multicoreware | 2019      |

## MEDIA

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|-----------------------------|-----------------------------------------------------------------|------|
| • Oregon State University   | AI student leverages OSU's interdisciplinary strengths          | 2025 |
| • Semiconductor Engineering | Grouping Complex Wafer Defect Patterns Into Meaningful Clusters | 2025 |
| • The Hindu                 | App for Plus Two Students                                       | 2015 |
| • Business Standard         | School Girl Develops App for TN Engg Aspirants                  | 2015 |
| • India.com                 | School Girl Develops Application for Tamil Nadu Engg Aspirants  | 2015 |

## SERVICE

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### Reviewer - Conference & Workshop

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|------------------------------------------------------------------------------------------|-----------|
| • ICML Workshop on Technical AI Governance Research (TAIGR)                              | 2026      |
| • ICLR Workshop on Generative AI in Genomics: Barriers and Frontiers (Gen <sup>2</sup> ) | 2026      |
| • AAAI Workshop on Artificial Intelligence with Biased or Scarce Data (AIBSD)            | 2026      |
| • AAAI International Conference on Web and Social Media (ICWSM)                          | 2025      |
| • ACL Workshop on DravidianLangTech                                                      | 2025      |
| • NeurIPS Workshop on Women in Machine Learning (WiML)                                   | 2024-2025 |

### Reviewer - Journal

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|--------------------------------------------------------------------------|------|
| • ACM Transactions on Knowledge Discovery from Data (TKDD)               | 2025 |
| • IEEE Transactions on Mobile Computing (TMC)                            | 2024 |
| • IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) | 2024 |

## CERTIFICATIONS

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2022 Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning - Coursera  
2022 Convolutional Neural Networks in TensorFlow - Coursera  
2022 Natural Language Processing in TensorFlow - Coursera  
2022 Introduction to Big Data - Coursera  
2022 Taming Big Data with Apache Spark and Python – Udemy  
2021 Amazon Web Services (AWS) - Machine Learning Specialty - Certified

## SKILLS

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- **Languages:** Python, R, C, C++, CUDA, MATLAB, SQL
- **A.I Tools:** TensorFlow, PyTorch, Diffusers, Transformers, OpenVINO, Keras, HuggingFace, Flower, Caffe, Scikit-learn, OpenCV, Kats, Pandas, NumPy, Jupyter, PySpark, Matplotlib, SciPy, Weights & Biases
- **R Libraries:** Dismo, Glmnet, Unmarked, DynamicSDM, Maxnet, Raster, Terra, Tidyverse, Nimble
- **Other Tools:** AWS (IoT Core, Sagemaker, Lambda, Kinesis, Glue, S3, SNS), Azure (IoT Hub, Stream Analytics, Function App), LaTeX, Git, Google Earth Engine, Docker